

Project title: Detection of helicobacter pylori in gastric biopsy

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1. Introduction

Helicobacter pylori (*H. pylori*) is a gram-negative, spiral-shaped bacterium that chronically colonizes the human gastric mucosa, infecting more than half of the world’s population. Classified by the World Health Organization and the International Agency for Research on Cancer as a Group I carcinogen, *H. pylori* is recognized as the primary risk factor for gastric cancer and a major contributor to various gastrointestinal diseases, including chronic gastritis, peptic ulcers, and mucosa-associated lymphoid tissue (MALT) lymphoma Teng et al., 2025; Klein et al., 2020. Its widespread prevalence and strong association with severe gastric disorders make it a significant global health concern.

The early and accurate detection of *H. pylori* is essential for preventing serious complications and reducing gastric cancer incidence through timely eradication therapy. Despite advances in diagnostic tools, histopathological examination of gastric biopsies remains the gold standard for identifying the bacterium. However, this process is highly labor-intensive, time-consuming, and prone to inter-observer variability, as it relies heavily on manual microscopic evaluation. The minute size of *H. pylori* and its morphological variability further complicate detection, often leading to missed diagnoses, particularly in cases with low bacterial density Backes, 2025. While alternative methods such as immunohistochemistry offer improved accuracy, their high cost and technical demands limit their widespread use, especially in resource-limited settings “CAD for *Helicobacter*” 2020.

Given these challenges, there is a growing need for more efficient, reliable, and accessible diagnostic solutions to support clinical decision-making and improve patient outcomes Zhou et al., 2020. Recent progress in digital pathology and computational technologies has opened new possibilities for enhancing diagnostic accuracy and reducing workload in pathology laboratories Liscia et al., 2022. Strengthening diagnostic capabilities in this area is crucial not only for improving individual patient care but also for addressing the broader global burden of *H. pylori* infection and its associated diseases.

This project focuses on the detection of *H. pylori* in gastric biopsy samples, aiming to contribute to more efficient, consistent, and accessible diagnostic practices. By improving the identification process of this clinically significant bacterium, the study supports the broader goal of reducing gastric cancer incidence and promoting better gastrointestinal health worldwide.

2. Literature review

Deep learning research into *H. pylori* detection addresses the challenges of large data scale and weak annotation through several distinct architectural paradigms.

2.1. Related Works

One approach is **Hybrid Hotspot Classification**, which employs a two-stage structure: coarse filtering of Whole Slide Images (WSIs) followed by deep learning classification on targeted regions of interest (ROIs). Klein et al., 2020 established an influential hybrid method-

ology designed to conserve computational resources by first isolating "HP hot spots" using conventional image processing (WSI downscaling, Otsu thresholding) before applying a compact VGG-style Convolutional Neural Network (CNN). This system achieved a critical clinical goal of 100% sensitivity for *HP* detection in Giemsa-stained sections, significantly surpassing traditional microscopic diagnosis sensitivity (68.4%). However, this high sensitivity came with a low specificity of 66.2% and a high False Positive Rate (FPR) of 41%. This performance shifted the diagnostic challenge from missing cases to generating excessive noise.

A second paradigm utilizes **Multi-Instance Learning (MIL)**, a weak supervision technique where the entire WSI is labeled, and the model must infer which individual patches contribute to the positive diagnosis. Teng et al., 2025 applied MIL models to H&E-stained WSIs to predict infection status. The DS-MIL model demonstrated superior performance, achieving an Accuracy of 89.7% and an Area Under the Curve (AUC) of 0.949, surpassing the baseline accuracy of senior pathologists (81.7%). Furthermore, the model achieved high sensitivity (94.3%), significantly outperforming pathologists (68.5% sensitivity). Further work on annotation ambiguity was conducted by Yang et al., who proposed a Weakly Supervised Multi-Task Learning Framework (WSMLF) for segmentation Yang et al., 2020. By extracting dual, contradictory targets from coarse polygon annotations, the system modeled label uncertainty, resulting in detection performance improvements of up to 6% (in F1 and IoU).

In data-scarce scenarios, reliance on deep CNNs pretrained on non-medical data has been questioned. Ibrahim et al., 2024 found that among various CNN baselines, ResNet-101 achieved the highest accuracy at 90.20%. Conversely, Backes, 2025 investigated conventional **texture descriptors** (e.g., Local Binary Patterns, Gray Level Dependence Matrix) combined through Particle Swarm Optimization (PSO) for feature selection. This low-complexity approach achieved an accuracy of 94.61% and an F1-score of 94.47% , surpassing the best CNN baseline (ResNet-101) by 4.41%. This outcome suggests that detection relies heavily on highly localized textural signals effectively captured by handcrafted features.

2.2. Limitations of Existing Studies

A rigorous assessment of precursor AI systems reveals critical limitations stemming from weak supervision and inadequate evaluation methods. The most significant flaw is **Annotation Inaccuracy**. The use of Active Learning (AL) failed to correct all errors, with approximately 10% of candidate regions remaining incorrectly labeled as negative, systematically limiting maximal attainable sensitivity. This issue is compounded by **Boundary Inconsistency**, where converting large, coarse WSI annotations into small patches contaminates the training set with false negatives if the strict patch boundary excludes crucial microbial content.

Compromised **Clinical Efficacy** often results from trade-offs in model design. Screening models, such as that by Klein et al. Klein et al., 2020, prioritized 100% sensitivity but tolerated a **High False Positive Burden** (41% FPR). This shifts the workload from diagnosis to manual validation of noise. Furthermore, **Human-AI Interaction Failure** was noted by Zhou et al. Zhou et al., 2020, where AI assistance increased diagnostic uncertainty on genuinely *HP*-negative cases. This led to a decrease in overall assisted accuracy, as uncertain diagnoses compelled costly ancillary testing (IHC/PCR) and delayed turnaround time (TAT).

Finally, **Methodological Flaws** hinder generalization. Standard bounding box metrics (IoU/DIoU) were found to be insufficient for evaluating tiny, sparse targets with coarse ground truth annotations. This metric mismatch renders localization metrics potentially unreliable "CAD for *Helicobacter*" 2020. Additionally, **Limited Generalization** is a major barrier, as many high-performing studies relied on specialized stains (Giemsa) or were confined to narrow single-center datasets. For example, Klein et al.'s model demonstrated poor robustness when deployed on

standard H&E (AUC 0.81) compared to Giemsa (AUC 0.92), failing a core requirement for commercial clinical deployment Klein et al., 2020.

3. Research Question, Hypothesis or Aim and Objectives

Can a computer-aided detection framework achieve high localization accuracy for minute, sparse bacterial targets while ensuring robust generalization across diverse, standard H&E staining protocols and mitigating the effects of weak label errors?

4. Research design/methodology

Our approach to detecting *Helicobacter pylori* in gastric biopsies addresses the challenge of analyzing extremely large images ($100k \times 50k$ pixels) where bacteria are sparsely distributed. Through our analysis, we've found that pylori bacteria consistently reside within gastric glands, which guides our detection strategy.

We start by segmenting gastric gland regions from the Whole Slide Images (WSIs). This preprocessing step dramatically reduces the search space, allowing our detection models to focus on areas where bacteria are most likely to be found rather than processing the entire tissue sample. This targeted approach improves both speed and accuracy.

After segmentation, we apply augmentation techniques to generate multiple variations of each tissue region. Our detection model analyzes these augmented versions, and a voting mechanism aggregates the predictions to produce a reliable final classification. This ensemble strategy helps us work effectively despite incomplete annotations in our training data.

A key insight is the asymmetric nature of detection: finding even one bacterium is enough to classify a sample as positive, so we can stop searching immediately once bacteria are detected. However, confirming a negative result requires thorough examination of all suspicious regions. To handle this efficiently, we propose a **hierarchical approach** with two models. A lightweight screening model quickly identifies potentially positive regions, then a more sophisticated model examines these areas in detail to confirm and precisely localize any bacteria present.

Regarding our training data, while we have a good number of samples, pathologists typically stopped annotating after finding the first bacterium, leaving many bacteria unmarked. We're addressing this through an **iterative refinement approach**: we'll train an initial model on existing annotations, then use it to detect additional unmarked bacteria in the same images. After expert review, these newly detected instances can be incorporated to progressively improve the model.

We're also exploring whether external datasets of pylori bacteria images can be augmented to visually match our LABOKLIN dataset. If successful, this would provide additional training data and improve model robustness.

5. Expected Outcome/ Implications and Contribution to the knowledge

We will deliver a functional system that veterinary clinicians can use in their diagnostic workflow. Users will upload whole slide images of gastric biopsies, and the system will automatically process them to detect the presence of pylori bacteria.

For positive cases, the system provides an annotated image with bounding boxes showing exactly where bacteria were detected. This visual localization helps pathologists quickly focus on relevant areas instead of manually searching through massive image files, significantly speeding up the diagnostic process. For negative cases, the system simply indicates no bacteria were found.

Our hierarchical approach creates a collaborative workflow where AI handles the time-consuming search task while pathologists focus on verification and clinical interpretation (Zhou et al., 2020; Liscia et al., 2022), ultimately accelerating the histopathological workflow and improving diagnostic consistency. Our contribution to knowledge lies in developing a novel, robust framework that mitigates the effects of weak supervision (labeling inconsistencies) and enhances generalization to standard H&E staining protocols.

6. Project plan

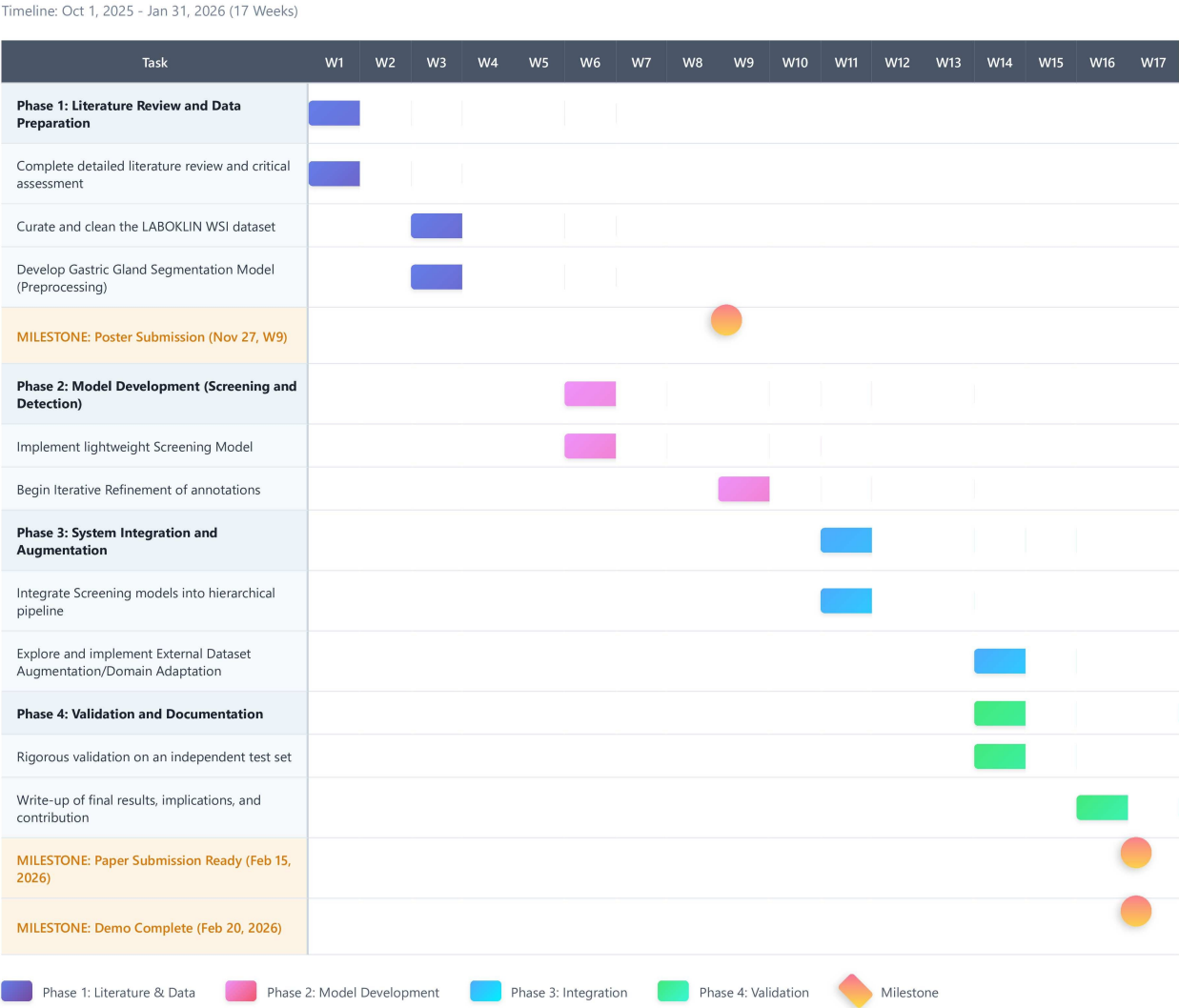


Figure 1: Project Deliverables Timeline

7. Ethical Consideration

The most critical ethical concern is **diagnostic accuracy** and its impact on animal welfare. False positives could lead to unnecessary treatments, significant financial costs, or even euthanasia decisions based on incorrect diagnoses. False negatives could result in untreated infections. We must rigorously validate the model before clinical deployment and position it strictly as a diagnostic aid that supports, never replaces, expert pathological judgment.

We will handle all veterinary medical images securely with appropriate data protection measures, ensuring compliance with relevant data privacy standards. The system will include clear documentation about its limitations, accuracy rates, and the requirement for professional verification of all findings before treatment decisions are made.

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